Day 117 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Use Base Area to Find Prism Volume

Standards: 6.G.A.2

Learning target: I can calculate prism volume with fractional edge lengths.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 19

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01 / Learn

A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

A rectangular prism has edges 4, 2 1/2, and 4 cm. Sketch it and find its volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Fractional edge lengths still multiply.

3. B = 4 × 5/2 = 10 cm²

4. V = B × 4 = 40 cm³.

5. State the result in the form requested: 40 cm³.

Answer: 40 cm³

Worked example 2

A prism measures 9/2 m by 3/2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 9/2 × 3/2 × 3/2 = 81/8.

3. The edge counts are 9, 3, 3, giving 81 cubes.

4. Each cube is 1/8 m³

5. 81/8 = 81/8.

Answer: 81/8 m³; 81 cubes

Day 117 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A rectangular prism has edges 7, 3 1/2, and 3 cm. Sketch it and find its volume.

02 / Constructed response

A rectangular prism has edges 7, 2, and 4 cm. Sketch it and find its volume.

03 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 3 cm. Sketch it and find its volume.

04 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 6 cm. Sketch it and find its volume.

05 / Constructed response

A rectangular prism has edges 2 1/2, 2 1/2, and 6 cm. Sketch it and find its volume.

06 / Constructed response

A rectangular prism has edges 3, 1, and 6 cm. Sketch it and find its volume.

07 / Constructed response

A rectangular prism has edges 4, 1, and 2 cm. Sketch it and find its volume. Show both lwh and Bh.

08 / Numeric entry and model reasoning

A prism measures 5/2 m by 3/2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Numeric entry and model reasoning

A prism measures 7/2 m by 3/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

10 / Constructed response

A rectangular prism has edges 2 1/2, 2, and 2 cm. Sketch it and find its volume. Show both lwh and Bh.

Day 117 | Unit 08 | Answer key

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1. 147/2 cm³

B = 7 × 7/2 = 49/2 cm²; V = B × 3 = 147/2 cm³.

6.G.A.2 | Constructed response

2. 56 cm³

B = 7 × 2 = 14 cm²; V = B × 4 = 56 cm³.

6.G.A.2 | Constructed response

3. 9 cm³

B = 2 × 3/2 = 3 cm²; V = B × 3 = 9 cm³.

6.G.A.2 | Constructed response

4. 18 cm³

B = 2 × 3/2 = 3 cm²; V = B × 6 = 18 cm³.

6.G.A.2 | Constructed response

5. 75/2 cm³

B = 5/2 × 5/2 = 25/4 cm²; V = B × 6 = 75/2 cm³.

6.G.A.2 | Constructed response

6. 18 cm³

B = 3 × 1 = 3 cm²; V = B × 6 = 18 cm³.

6.G.A.2 | Constructed response

7. 8 cm³

B = 4 × 1 = 4 cm²; V = B × 2 = 8 cm³.

6.G.A.2 | Constructed response

8. 15/2 m³; 60 cubes

Multiply edges: 5/2 × 3/2 × 2 = 15/2. The edge counts are 5, 3, 4, giving 60 cubes. Each cube is 1/8 m³; 60/8 = 15/2.

6.G.A.2 | Numeric entry and model reasoning

9. 21/4 m³; 42 cubes

Multiply edges: 7/2 × 3/2 × 1 = 21/4. The edge counts are 7, 3, 2, giving 42 cubes. Each cube is 1/8 m³; 42/8 = 21/4.

6.G.A.2 | Numeric entry and model reasoning

10. 10 cm³

B = 5/2 × 2 = 5 cm²; V = B × 2 = 10 cm³.

6.G.A.2 | Constructed response